12/8/2025
NKU RT – Liquids

Making of the Rocket Team website

Jordan Niemann, Aaron Milner

Introduction

While not ultra-exclusive to the liquid rocket engine project, this is a guide crafted by Jordan Niemann for Aaron Milner on how the website for Rocket Team shall be created.

Inspirations for website design

The following rocketry organization websites are great reference, both for design and content:

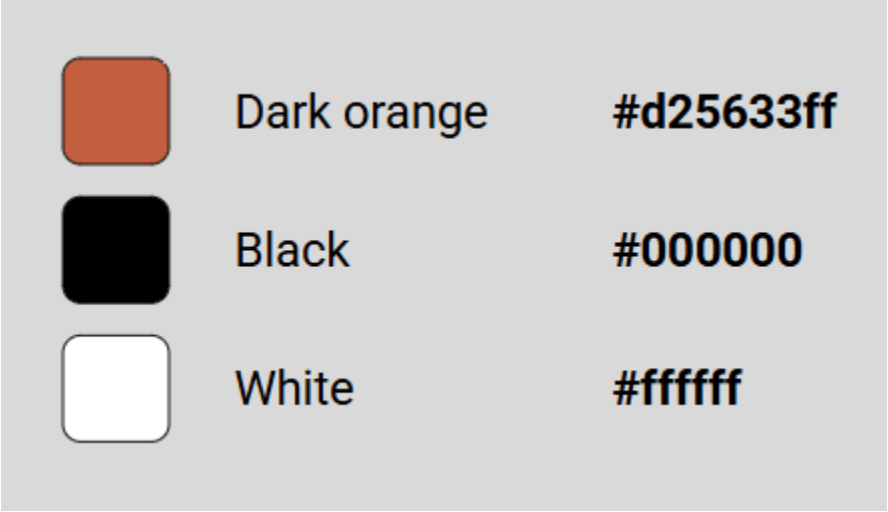
- https://purpl.space/
- https://purdueseds.space/
- https://www.berkeleyse.org/
- https://www.tamured.space/
- https://texasrocketlab.ae.utexas.edu/
- https://www.gtspaceprogram.com/

Color palette & logo

The current logo for the rocket team is provided below.



The primary colors for the website, including hex codes, are:



Sections

There shall be multiple sections of the website, organized in an easy-to-navigate way either on the top of the screen or with a pull-out sidebar (see websites referenced in *Inspirations for website design* for an example).

The questions, as of the making of the website, shall be as follows:

I. **Home**
II. **People**
III. **Projects**
    A. **Star Catcher**
    B. **Liquid rocket engine**
IV. **Contact us**
V. **Sponsors**

Each section and its contents are outlined below in subsections.

Home

This can really have anything. Usually it'll have some cool videos, some pictures of the team doing stuff, some stats (which we can't really have because our numbers aren't that impressive yet). Sometimes it'll have what is a few of the sections on the home page (like in this example) or it won't have much at all (like in this example).

Cheesy catchphrase under the name (see example on this home page): **Bringing liquid rocketry to Northern Kentucky.**

Short description of the team with a cheesy heading on top, such as "Our goal" or "Who we are": **NKU Rocket Team was founded to provide students hands-on engineering experience in design, manufacturing, testing and launching through solid and liquid rocketry projects. With an emphasis on designing all rockets and engines from scratch (excluding solid rocket motors), NKU Rocket Team takes pride in building critical skills and instilling a first principles mindset within its members.**

Some cool pictures that can be used are here:







People

This section would theoretically be where all of the e-board members go. As of right now, I haven't asked anyone for permission to post their pictures and I'm not sure of the pictures would be super high-quality as of right now, so I say this section can be put on hold.

Projects

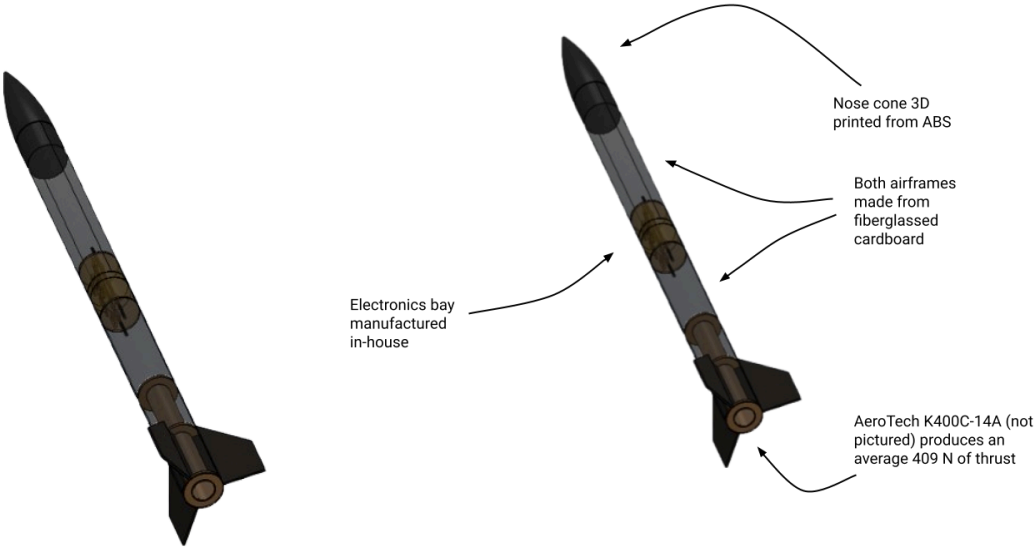
Star Catcher

Some nice text to start it off: **NKU Rocket Team's first solid rocketry project will reach over 5,000 ft of altitude and is slated to launch in January of 2026.**

Stats in a cool-looking table (like in this page):

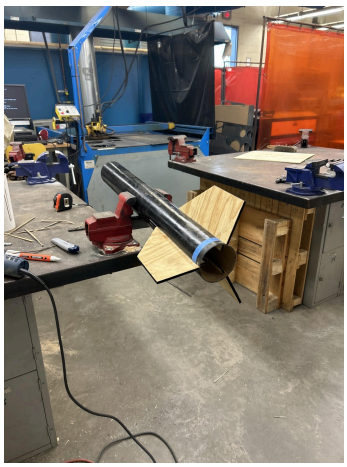
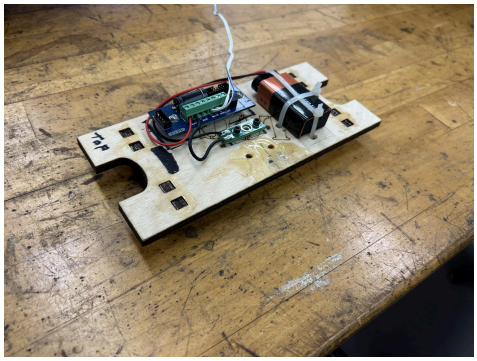
- **Body diameter: 4 in**
- **Total length: 72 in**
- **Motor choice: AeroTech K400C-14A**
- **Average thrust: 409 N**
- **Total impulse: 1,304 N-s**
- **Expected apogee: 5,191 ft**

Cool interactive rocket CAD model (like in this page):



Some cool pictures of the rocket at various stages of manufacturing, with some captions, maybe in some sort of image carousel:

Picture 1 – **Our nose cone was printed in two pieces due to its length, and used ABS as a filament. Pictured is the top piece of the nose cone immediately after printing.**
Picture 2 – **The rocket, held in a vise, directly after fin attachment.**
Picture 3 – **The electronics bay consisted of an altimeter sourced off-the-shelf, a battery and a screw switch. The plywood was laser cut by the manufacturing team.**

Liquid rocket engine

Some nice text to start it off: **The first student-built liquid rocket engine in the history of the Cincinnati area.**

Stats in a cool-looking table (like in this page):

- **Thrust: 300 lbf**
- **Propellants: Isopropyl alcohol and nitrous oxide**
- **Cycle: Pressure-fed**
- **Cooling: Heatsink**
- **Injector type: Pintle**

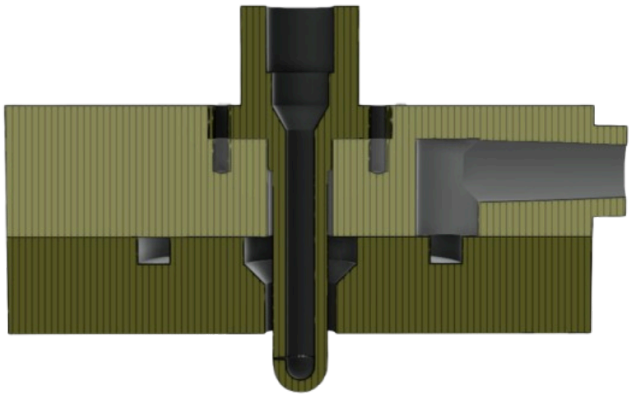
Section about project goals:

This shall be NKU Rocket Team's first-ever liquid rocket engine, with an aim of building skills within members to build more advanced engines in the future. Nitrous oxide is used as an oxidizer due to its self-pressurizing capability and non-cryogenic nature, which simplifies the feed system greatly. The project is on an accelerated timeline, with an aim to hotfire by May of 2026.

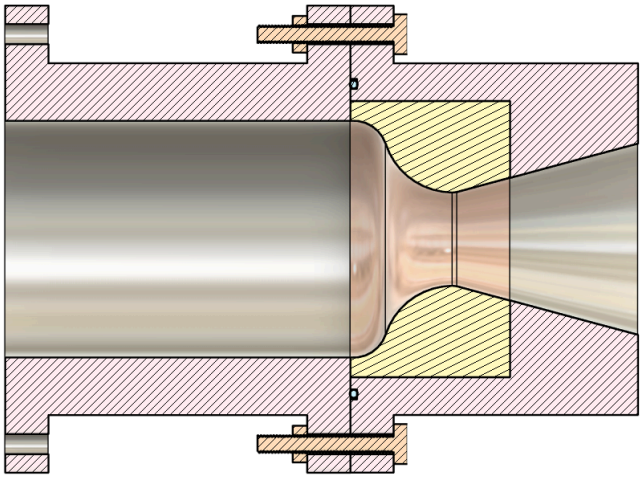
Section about design:

All engine components are designed from scratch, including the injector and thrust chamber, in Siemens NX and Creo, with Autodesk Fusion used for half-sections. This builds critical skills that are needed for industry.

Some good images for this section are below:



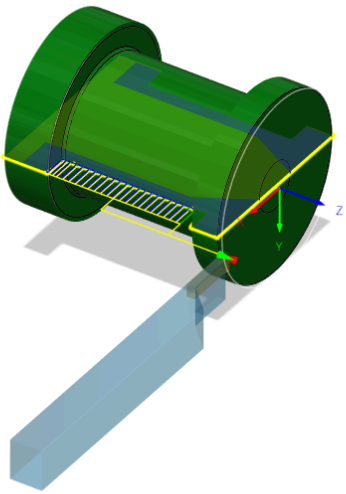


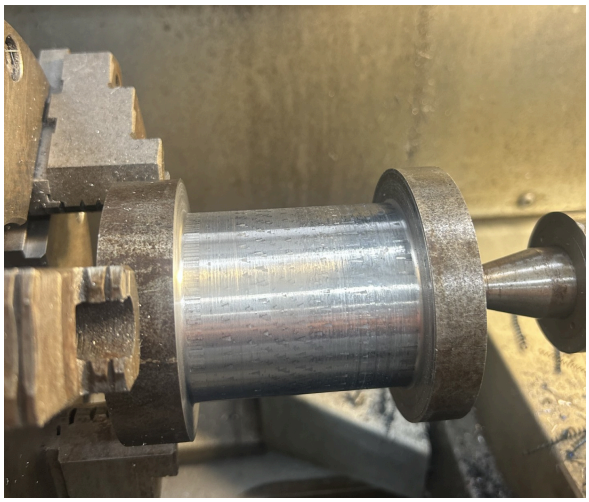


Section about manufacturing:

All manufacturing is done in-house at NKU's machine shop. Students work closely with the machine shop director to machine and assemble critical engine components. Fusion CAM is used for programming CNC machines. Demonstrations of manufacturing are done using shop-supplied steel to ensure that the available machines and tools can provide the geometries required.

Some good images for this section are below:





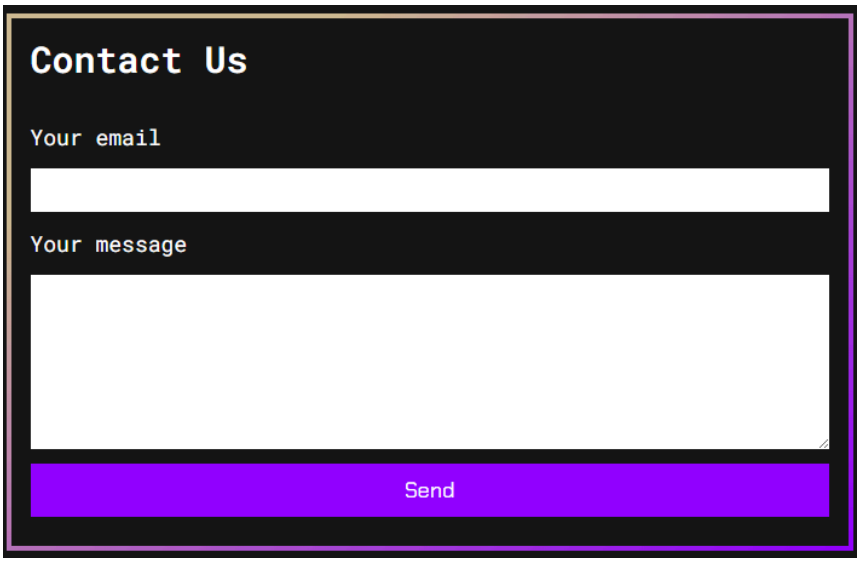




Contact us

This page was used as an inspiration for this.

For anyone interested in joining or supporting the club, we would love to talk! We can be contacted using the form below (insert a form like the one below):



Email: niemannj2@mymail.nku.edu
LinkedIn: nku-rt

Sponsors

Nice message at the top: **We are currently seeking sponsors to help support our solid and liquid rocketry projects! If you would like support us, please reach out to the following people:**

President

Jordan Niemann – niemannj2@mymail.nku.edu

Faculty mentor

Dr. Dirk Grupe – gruped1@nku.edu